



Question 1

The Gibbs definition of entropy is:

$$S = -k_B \sum_s p_s \lg p_s$$

$$p_s = \frac{e^{\beta \mu N_s - \beta E_s}}{\mathcal{Z}}$$

$\mathcal{Z} \rightarrow$ Grand Partition Function
 $\mathcal{Z} = \sum_s e^{\beta \mu N_s - \beta E_s}$

so that

$$S = -k_B \sum_s \frac{e^{\beta \mu N_s - \beta E_s}}{\mathcal{Z}} \left[\beta \mu N_s - \beta E_s - \lg \mathcal{Z} \right]$$

$$= -k_B \beta \mu \underbrace{\sum_s \frac{e^{\beta \mu N_s - \beta E_s}}{\mathcal{Z}} N_s}_{\langle N \rangle} + k_B \beta \underbrace{\sum_s \frac{e^{\beta \mu N_s - \beta E_s}}{\mathcal{Z}} E_s}_{\langle E \rangle} + \underbrace{\sum_s \frac{e^{\beta \mu N_s - \beta E_s}}{\mathcal{Z}} \lg \mathcal{Z}}_{\lg \mathcal{Z}}$$

$$= -\frac{\mu}{T} \langle N \rangle + \frac{\langle E \rangle}{T} + \lg \mathcal{Z}$$

Recall that $-\lg \mathcal{Z} \cdot \beta^{-1} = \Phi = -PV$

so that $\lg \mathcal{Z} = PV \cdot \beta = \frac{PV}{k_B T}$ which for an ideal gas is $\langle N \rangle$

Therefore for an ideal gas:

$$S = -\frac{\mu}{T} \langle N \rangle + \frac{\langle E \rangle}{T} + \langle N \rangle k_B$$

note that in the ideal gas $\langle E \rangle = \langle N \rangle \cdot \frac{3}{2} k_B T$

Therefore
$$S = -\frac{\mu}{T} \langle N \rangle + \frac{3}{2} \frac{k_B T}{T} \langle N \rangle + \langle N \rangle k_B = -\frac{\mu}{T} \langle N \rangle + \frac{5}{2} k_B \langle N \rangle$$

as a Result:

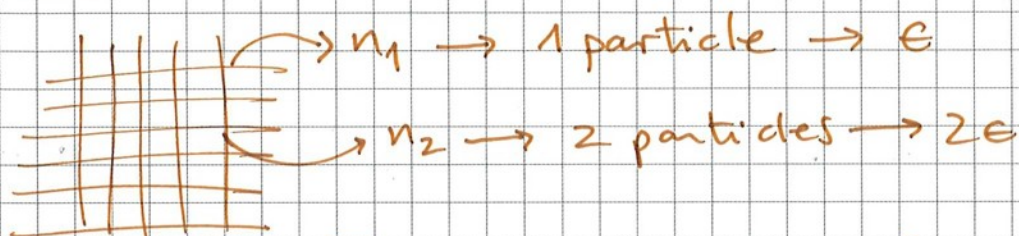
$$S = k_B \langle N \rangle \left(-\beta\mu + \frac{\epsilon}{2} \right)$$

since $S \geq 0$

$$\beta\mu \leq \frac{\epsilon}{2} \quad \text{or} \quad \mu \leq \frac{\epsilon}{2} k_B T$$



Problem 1:



$$\text{Energy} = n_1 \cdot E + n_2 \cdot 2E$$

where $n_1 + n_2 = N$ (all sites are occupied)

$$\text{Energy} = n_1 E + (N - n_1) 2E = 2EN - n_1 E = E(2N - n_1)$$

fixed E implies fixed $n_1 \rightarrow$ the number of possible choices is

$$\Omega = \binom{N}{n_1} \rightarrow S = k_B \log \binom{N}{n_1}$$

in the thermodynamic limit $S = k_B [N \log N - N \log n_1 - (N - n_1) \log (N - n_1)]$
(N, n_1 large)

In terms of $x_1 \equiv \frac{n_1}{N}$ after some algebra we get

$$\begin{aligned} a) \quad S &= k_B N [\log N - x_1 \log n_1 - (1 - x_1) \log (N - n_1)] \\ &= k_B N [\log N - x_1 (\log \frac{N}{N} + \log N) - (1 - x_1) (\log N + \log (1 - x_1))] \\ &= -k_B N [x_1 \log x_1 + (1 - x_1) \log (1 - x_1)] \end{aligned}$$

b) To compute the dependency of x_1 in E we use

$$\left(\frac{\partial S}{\partial E} \right)_V = \frac{1}{T} = \frac{\partial S}{\partial x_1} \frac{dx_1}{dE} = \frac{\partial S}{\partial x_1} \cdot \frac{1}{\frac{dE}{dx_1}}$$

$$\frac{dE}{dx_1} = \frac{dE}{dn_1} \cdot \frac{dn_1}{dx_1} = -e \cdot N$$

$$\frac{\partial S}{\partial E} = -e \cdot N \cdot \frac{\partial S}{\partial x_1} = (-e \cdot N)^{-1} \cdot (-k_B N) \cdot \left[x_1 \frac{1}{x_1} + \log x_1 - \frac{1-x_1}{1-x_1} - \log(1-x_1) \right]$$

$\rightarrow n_1 = x_1 \cdot N$

$$\frac{\partial S}{\partial E} = \frac{k_B N}{e \cdot N} \cdot \log \frac{x_1}{1-x_1} = \frac{1}{T}$$

$$\frac{x_1}{1-x_1} = e^{E/k_B T}$$

$$x_1(1+e^{E/k_B T}) = e^{E/k_B T}$$

$$\boxed{x_1 = \frac{e^{E/k_B T}}{1+e^{E/k_B T}} = \frac{1}{1+e^{-E/k_B T}}}$$

$x_1 \text{ max} \rightarrow T=0$ since the min energy is $E = N \cdot e$

let's check:

$$\lim_{T \rightarrow 0} x_1 = \frac{1}{1+e^{-E/k_B T}} \rightarrow 1$$

$x_1 \text{ min} \rightarrow T \rightarrow \infty$

$$x_1 = \lim_{T \rightarrow \infty} \frac{1}{1+e^{-E/k_B T}} = 1/2$$

Therefore x_1 goes is a value in the range $[0, 1/2]$ and not $[0, 1]$ because at high temperatures half of the sites have one part and half of the sites have 2 particles.



Problem 2: Gas of fermions in 2D.

To get the density of states in a 2D system of surface $S = L^2$ we do as usual: ~~use~~

$$\sum_{\substack{\vec{n} + \text{deg} \\ \text{quantum number}}} \rightarrow \int d\epsilon g(\epsilon)$$

We need to find what is the number of states comprised in an energy slice $(\epsilon, \epsilon + d\epsilon)$

The energy of a fermion in a gas: $\epsilon_i = \frac{\hbar^2 k^2}{2m_e}$

with $k_x = \frac{\pi n_x}{L}$ $k_y = \frac{\pi n_y}{L}$

in polar coordinates $k_x^2 + k_y^2 = \frac{\pi^2}{L^2} n^2$

So that $\int_0^\infty dk_y \int_0^\infty dk_x \rightarrow 4 \int_0^\infty \int_0^\infty dk_x dk_y \rightarrow 4 \int_0^\infty dn \cdot \int_0^{2\pi} d\phi$

polar coordinates
 n, ϕ

$dx dy = r dr d\phi$

$$= \frac{2\pi}{4} \int_0^\infty dn \cdot n \rightarrow \left\{ \begin{array}{l} k = \frac{\pi}{L} n \\ \epsilon = \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L} \right)^2 n^2 \end{array} \right. \quad \boxed{d\epsilon = \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L} \right)^2 2n dn}$$

$$n dn = \left(\frac{L}{\pi} \right)^2 \frac{2m_e}{\hbar^2} d\epsilon$$

$$= \frac{2\pi}{4} \left(\frac{L}{\pi} \right)^2 \frac{2m_e}{\hbar^2} \int_0^\infty d\epsilon$$

Finally because we have fermions with $s = \frac{1}{2}$ we have a degeneracy of 2 states per each unique set of quantum numbers.

$$\sum_{\substack{\vec{n} + \text{deg} \\ \text{quantum number}}} \rightarrow \int d\epsilon g(\epsilon)$$

$$= \int_0^\infty d\epsilon \cdot 2 \cdot \frac{2\pi}{4} \left(\frac{L}{\pi} \right)^2 \frac{2m_e}{\hbar^2}$$

$g(\epsilon) = \frac{L^2}{\pi} \frac{2m_e}{\hbar^2} = \frac{2S m_e}{\pi \hbar^2}$

b) Chemical potential.

we have N atoms that give each one a free- e^- ,
therefore, we have N electrons.

The number of electrons has to be equal to the total occupancy of all states:

$$N = \int_0^{\infty} d\epsilon \, g(\epsilon) \cdot n(\epsilon)$$

in fermions $n(\epsilon) = \frac{e^{-\beta(\epsilon-\mu)}}{1+e^{-\beta(\epsilon-\mu)}}$

$$N = \int_0^{\infty} d\epsilon \, \frac{2Sm_e}{\pi \hbar^2} \cdot \frac{e^{-\beta(\epsilon-\mu)}}{1+e^{-\beta(\epsilon-\mu)}}$$

the integrand has $e^{-\beta(\epsilon-\mu)}$ we change the integration variable $\epsilon \rightarrow x = \beta(\epsilon-\mu)$

so that $dx = \beta d\epsilon$ and the limits are $\left\{ \begin{array}{l} \epsilon=0 \quad x=-\beta\mu \\ \epsilon \rightarrow \infty \quad x \rightarrow \infty \end{array} \right.$ ($\beta \neq 0$)

$$N = \int_{-\beta\mu}^{\infty} dx \, \frac{2Sm_e}{\pi \hbar^2} \frac{1}{\beta} \left[\frac{e^{-x}}{1+e^{-x}} \right]$$

$$= \frac{2Sm_e}{\pi \hbar^2} \frac{1}{\beta} \left[-\frac{d}{dx} \log(1+e^{-x}) \right]_{-\beta\mu}^{\infty} = \frac{2Sm_e}{\pi \hbar^2} \frac{1}{\beta} \log(1+e^{\beta\mu})$$

So that $N = \frac{2Sm_e}{\pi \hbar^2} \frac{1}{\beta} \log(1+e^{\beta\mu})$

$$\rightarrow 1+e^{\beta\mu} = e^{\beta N \pi \hbar^2 / 2Sm_e}$$

$$\mu = k_B T \log(e^{\beta N \pi \hbar^2 / 2Sm_e} - 1)$$

Fermi energy $\mu(T=0)$ $E_F = \lim_{T \rightarrow 0} \mu = k_B T \frac{\beta N \pi \hbar^2}{2Sm_e} = \frac{N \pi \hbar^2}{2Sm_e}$ ✓

Done directly at $T=0$: $N = \int_0^{E_F} g(\epsilon) d\epsilon = \frac{2Sm_e}{\pi \hbar^2} E_F$ $E_F = \frac{N \pi \hbar^2}{2Sm_e}$ ✓